

People's Democratic Republic of Algeria

Ministry of Higher Education and Scientific Research

Numidia Institute of Technology

Transformation and Analysis of a PCAP-based Malicious API Traffic Dataset

Internship Report

Students:

BENNOUAR Tarek

HANI Nour El Houda

Host Organization:

CERIST (Centre de Recherche sur l'Information Scientifique et Technique)

Supervisors:

Dr. Nouali Omar

Dr. Amina Djellalbia

Academic Year 2024–2025

Abstract

This report presents the internship project carried out at the Centre de Recherche sur l'Information Scientifique et Technique (CERIST). The work focuses on the transformation and exploratory analysis of a raw and fragmented network traffic dataset containing malicious API requests. Starting from poorly structured PCAP files, a complete data processing pipeline was designed and implemented, including data extraction, cleaning, normalization, dataset fusion, and deduplication. The processed data was further enriched through log simulation and subjected to an extensive exploratory data analysis (EDA). The objective of this work was to produce a unified and high-quality dataset suitable for cybersecurity research, particularly for the detection of API-based attacks. The results highlight both the strengths and the inherent limitations of the dataset and provide recommendations for future improvements and research directions.

Acknowledgements

I would like to express my sincere gratitude to the CERIST research team for their support and guidance throughout the internship period. Special thanks are addressed to our supervisors, Dr. Nouali Omar and Dr. Djellabia Amine for their valuable technical advice and continuous encouragement. I also thank my academic advisors, Dr. Guenfaf Youcef and Dr. Mazouni Fares, for their constructive feedback and academic supervision. Finally, I extend my appreciation to all colleagues and researchers at CERIST who contributed to a productive and stimulating research environment.

Contents

Abstract	i
Acknowledgements	ii
1 Introduction	1
1.1 Context and Motivation	1
1.2 Objectives of the Internship	1
2 Host Organization	2
2.1 CERIST Overview	2
2.2 Research Environment	2
3 Methodology	3
3.1 Data Extraction from PCAP Files	3
3.2 Normalization and Cleaning	3
3.3 Dataset Fusion and Deduplication	3
3.4 Log Simulation	4
3.5 Exploratory Data Analysis	4
4 Results and Discussion	5
4.1 Results and Discussion	5
4.1.1 Dataset Characteristics	5
4.2 Biases and Limitations	11
4.3 Implications for Security Research	12
5 Skills and Lessons Learned	13
5.1 Technical Skills	13
5.2 Analytical and Professional Skills	13
6 Conclusion and Perspectives	14

Chapter 1

Introduction

1.1 Context and Motivation

The rapid adoption of web Application Programming Interfaces (APIs) has profoundly transformed modern software architectures. APIs now constitute a critical component of web applications, cloud services, and microservice-based systems. However, this evolution has also introduced new security challenges, as attackers increasingly target API endpoints using crafted HTTP requests to exploit vulnerabilities such as SQL injection, cross-site scripting, or local file inclusion. In this context, high-quality and realistic datasets are essential for developing, evaluating, and benchmarking security detection mechanisms, as highlighted in industry references such as the OWASP API Security Top 10 [6].

1.2 Objectives of the Internship

The primary objective of this internship was to transform a collection of unstructured and attack-specific PCAP files into a unified, clean, and analysis-ready dataset suitable for cybersecurity research. This objective was achieved through the automation of data extraction, cleaning, normalization, merging, and deduplication processes. A secondary objective was to simulate realistic web server logs and to conduct an in-depth exploratory data analysis in order to assess dataset quality, identify biases, and characterize attack patterns.

Chapter 2

Host Organization

2.1 CERIST Overview

The Centre de Recherche sur l'Information Scientifique et Technique (CERIST) is a major Algerian public research institution operating under the Ministry of Higher Education and Scientific Research. Established in 1985, CERIST played a pioneering role in the development of Internet infrastructure in Algeria and has been the official registry of the national “.dz” domain since 1994. Its missions include the development of national scientific information systems, the promotion of research in information and communication technologies, and the support of digital transformation initiatives.[2]

2.2 Research Environment

CERIST provides a multidisciplinary research environment with a strong focus on network infrastructure, information systems, and cybersecurity. The internship was conducted within a research-oriented setting, offering access to technical expertise, datasets, and computational resources relevant to security analysis and network traffic processing.

Chapter 3

Methodology

3.1 Data Extraction from PCAP Files

The initial dataset consisted of multiple PCAP files, each corresponding to a specific type of web attack. These files were highly unstructured and not directly usable for analysis. Python scripts based on PyShark and PcapKit were developed to extract HTTP requests and responses from the PCAP files and convert them into enriched JSON representations containing relevant fields such as HTTP method, URL, headers, request body, and response content [4, 7].

3.2 Normalization and Cleaning

The extracted data exhibited inconsistencies and missing fields due to heterogeneous capture conditions. A normalization process was applied to standardize JSON structures, reconstruct incomplete URLs, normalize HTTP methods, and ensure uniform header representation. Non-informative fields were removed, and parsing errors were corrected, resulting in a clean and homogeneous dataset.

3.3 Dataset Fusion and Deduplication

All normalized JSON files were merged into a single comprehensive dataset containing samples from multiple attack categories. Exact duplicates were identified and removed based on key request and response attributes, ensuring dataset uniqueness. The final dataset was shuffled to interleave attack types and avoid mono-class blocks.

3.4 Log Simulation

To improve realism and compatibility with security monitoring tools, the dataset was transformed into a web server log format inspired by standard NGINX access logs [5]. Additionally, the data was flattened into a tabular CSV format to facilitate statistical analysis, visualization, and machine learning workflows.

3.5 Exploratory Data Analysis

A comprehensive exploratory data analysis was conducted to assess dataset quality and uncover structural patterns. The interpretation of HTTP methods and status codes follows standard HTTP semantics as defined in the corresponding specifications [3].

The complete source code implementing the data extraction, preprocessing, log simulation, and exploratory analysis pipeline is publicly available for reproducibility purposes [1].

Chapter 4

Results and Discussion

4.1 Results and Discussion

This section presents the main characteristics of the constructed dataset and discusses the insights obtained through exploratory data analysis. The objective is to assess the quality, representativeness, and limitations of the dataset from a cybersecurity research perspective.

4.1.1 Dataset Characteristics

Distribution of Attack Types

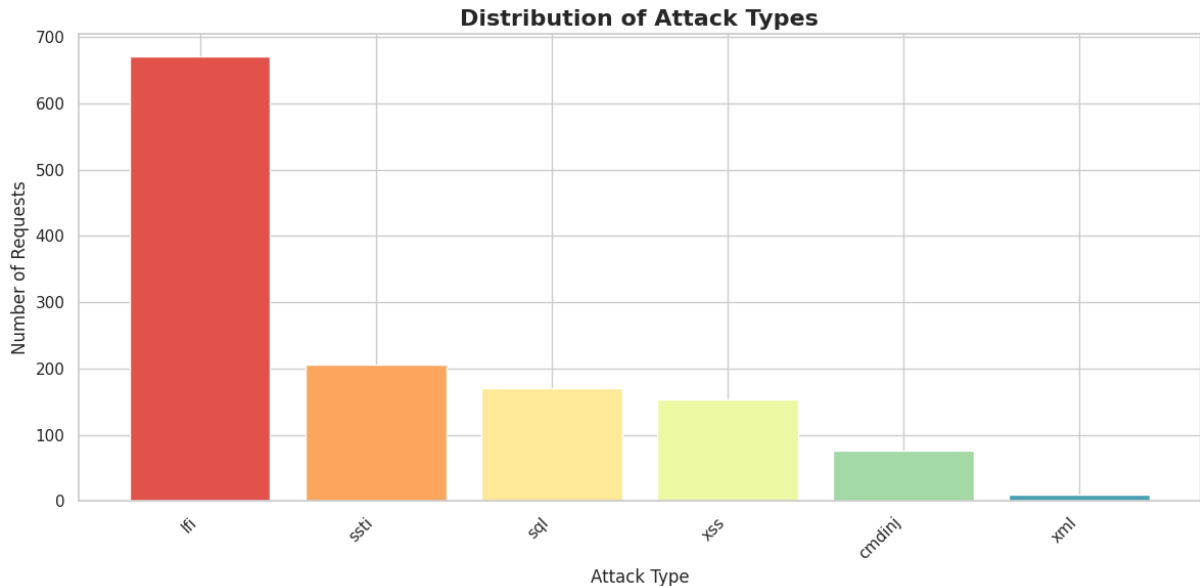


Figure 4.1: Distribution of attack types in the dataset

illustrates the distribution of attack categories within the dataset. A pronounced class imbalance is immediately observable, revealing fundamental structural characteristics of

the dataset. Local File Inclusion (LFI) attacks constitute the overwhelming majority, accounting for nearly 1,500 requests. In contrast, SQL Injection (SQLi) and Cross-Site Scripting (XSS) attacks are significantly underrepresented, with approximately 200 requests each. Server-Side Template Injection (SSTI) attacks appear even less frequently, while XML-based attacks are virtually absent.

This imbalance reflects the composition of the original PCAP sources rather than realistic attack distributions in production environments. From a machine learning perspective, this structure introduces a strong bias toward dominant classes and requires careful evaluation using per-class metrics rather than overall accuracy.

HTTP Methods and Status Code Distribution

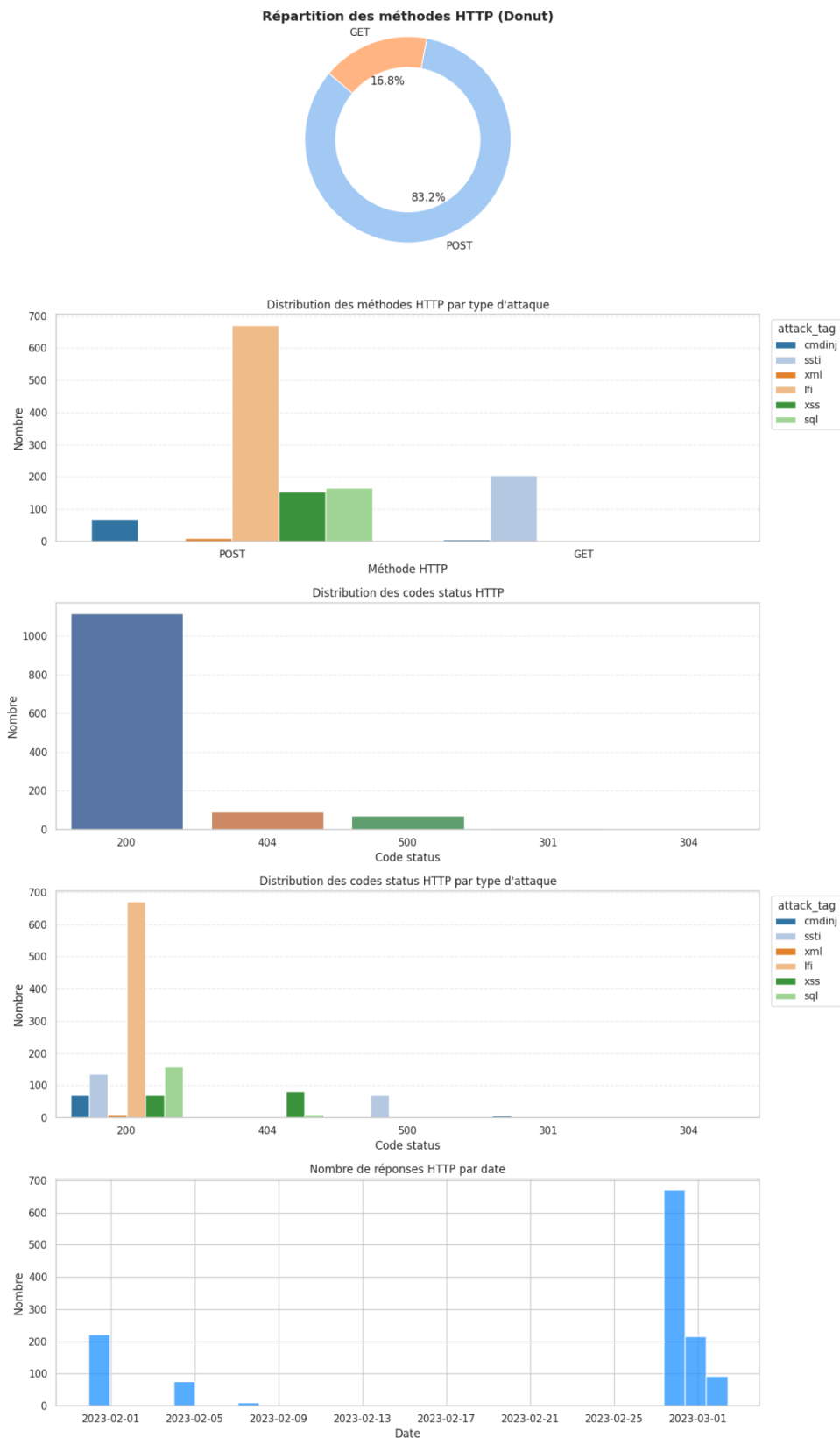


Figure 4.2: Distribution of HTTP methods and status codes

presents the distribution of HTTP methods and response status codes observed in the dataset. POST requests dominate the traffic, representing more than 80% of all samples, while GET requests account for a significantly smaller proportion. This behavior is consistent with modern API attack strategies, where malicious payloads are primarily delivered through request bodies.

The distribution of HTTP status codes is heavily skewed toward status code 200 (OK), indicating that most malicious requests were successfully processed by the target applications. Error codes such as 404 and 500 are present but marginal, suggesting a controlled or permissive execution environment rather than realistic production conditions.

Request Body Presence Across Attacks

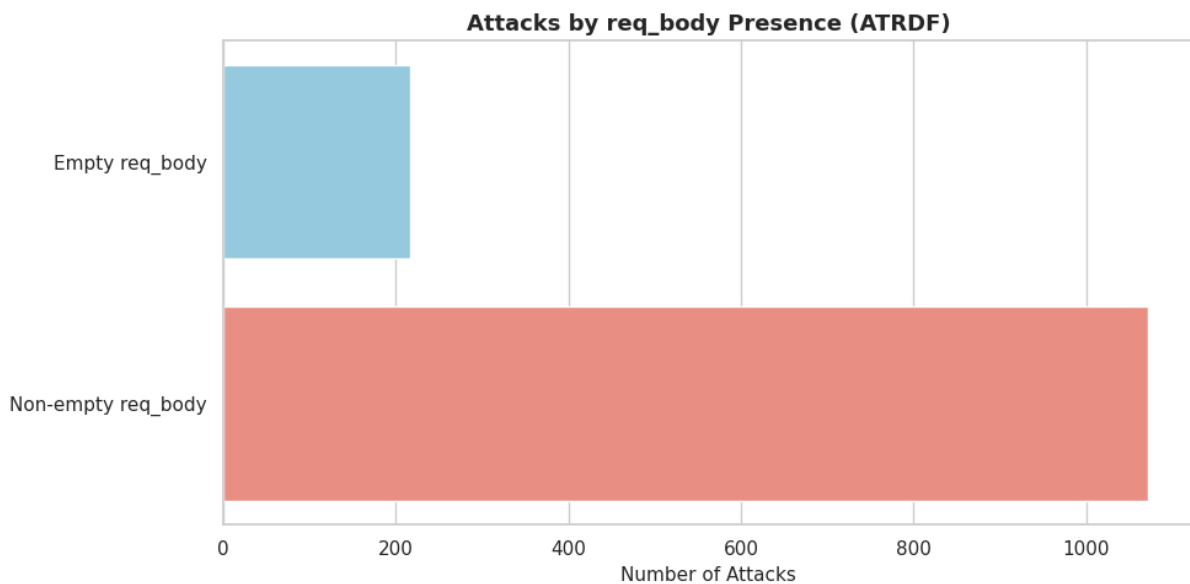


Figure 4.3: Distribution of attacks by request body presence

analyzes the presence of request bodies in malicious requests. The majority of attacks contain a non-empty request body, confirming the dominance of POST-based attack techniques. This observation aligns with the prevalence of injection-style attacks that rely on body-embedded payloads.

Nevertheless, a smaller fraction of attacks operate without a request body, relying instead on URL parameters. This highlights the coexistence of multiple attack delivery mechanisms and implies that effective detection systems must analyze both request bodies and URL structures.

URL Length and Structural Complexity

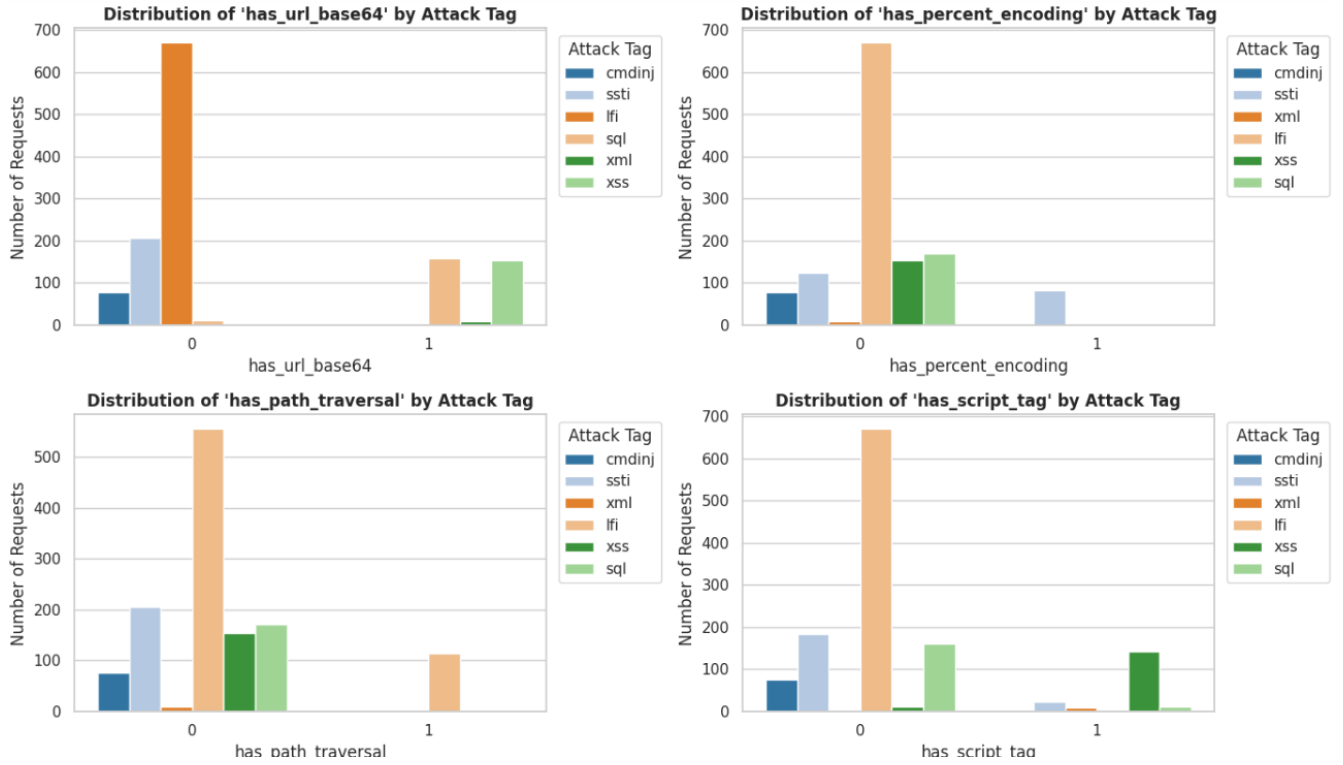


Figure 4.4: URL length distribution and attack-specific patterns

presents the distribution of URL lengths within the dataset. The distribution is strongly right-skewed, with most URLs remaining relatively short while a limited number of outliers exceed several hundred characters. These long URLs typically correspond to heavily parameterized or encoded payloads.

When analyzed by attack category, distinct patterns emerge. SQL injection attacks tend to generate longer URLs due to complex parameter manipulation, whereas LFI attacks generally exhibit moderate lengths reflecting file path traversal attempts. XSS attacks display the widest variance, indicating heterogeneous payload structures. These differences suggest that URL length constitutes a potentially informative feature for attack classification.

URL Keyword and Parameter Frequency Analysis

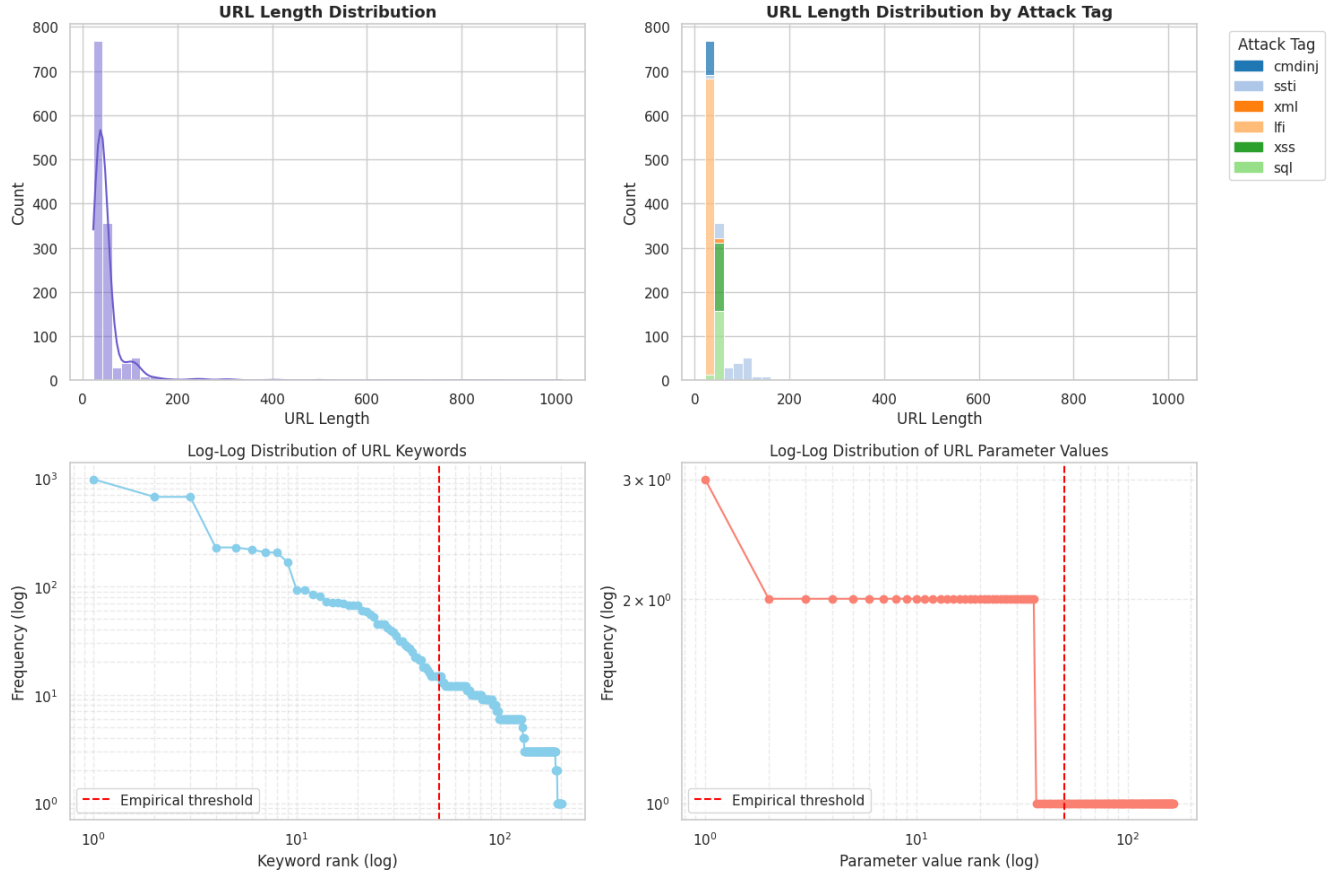


Figure 4.5: Log-log distribution of URL keywords and parameter values

shows the log-log distributions of URL keywords and parameter values. Both distributions exhibit heavy-tailed behavior characteristic of Zipf-like laws, where a small number of tokens appear very frequently while the majority occur rarely.

This pattern reveals the presence of recurring attack signatures and commonly reused payload components. While such regularities can be effectively exploited by signature-based detection systems, they also highlight a limitation of the dataset, as rare or novel payload variations are underrepresented.

User-Agent Distribution

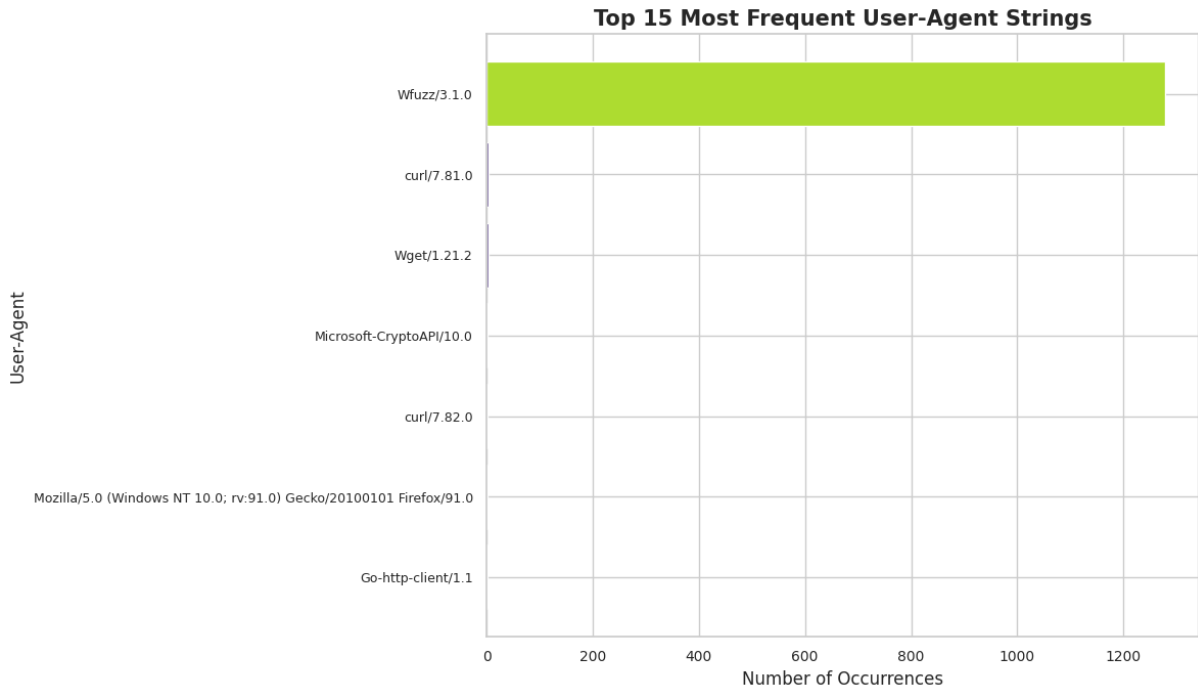


Figure 4.6: Top 15 most frequent User-Agent strings

presents the most frequent User-Agent strings observed in the dataset. The traffic is overwhelmingly dominated by automated tools, particularly *Wfuzz/3.1.0*, followed by command-line utilities such as *curl* and *Wget*. Legitimate browser-based User-Agents are almost entirely absent.

This lack of diversity emphasizes the synthetic and tool-driven nature of the dataset. Although this characteristic simplifies attack labeling and analysis, it limits the representativeness of the dataset for real-world API traffic, where benign and malicious requests are typically interleaved.

4.2 Biases and Limitations

Several limitations were identified during the analysis. The dataset is affected by significant class imbalance, with certain attack types being overrepresented. The dominance of automated tools results in low user-agent diversity, and homogeneous server responses limit the applicability of the dataset for anomaly-based detection approaches. Moreover, the strong correlation between attack features and labels may lead to overfitting in machine learning models.

4.3 Implications for Security Research

Despite the identified limitations, the constructed dataset remains a valuable resource for controlled cybersecurity research. Its clear labeling, internal consistency, and structured format make it well suited for benchmarking signature-based and supervised machine learning models.

However, results obtained using this dataset should not be directly extrapolated to real-world environments without additional validation. Future work should focus on balancing attack classes, integrating benign traffic, and increasing diversity in User-Agent strings, payloads, and server responses to improve realism and robustness.

Additional exploratory analyses, including URL structure, request body presence, and attack-specific binary features, are provided in Appendix A for completeness.

Chapter 5

Skills and Lessons Learned

5.1 Technical Skills

The internship enabled the development of strong technical skills in network traffic analysis, PCAP processing, advanced Python scripting, data normalization and cleaning, dataset engineering, and log simulation.

5.2 Analytical and Professional Skills

In addition to technical expertise, the project strengthened analytical skills related to data visualization, statistical analysis, and critical evaluation of dataset quality. The internship also contributed to professional development through project management, technical documentation, and collaboration in a research environment.

Chapter 6

Conclusion and Perspectives

This internship bridged the gap between raw network captures and actionable cybersecurity analytics by producing a structured dataset for the study of malicious API traffic. While the work highlights the value of carefully processed datasets, it also emphasizes the need for realism and diversity in security research. Future work should focus on enriching the dataset with additional attack types, integrating benign traffic, and simulating more heterogeneous environments to enhance research applicability.

Bibliography

- [1] Bennouar, Tarek and Hani, Nour El Houda. Cerist http malicious dataset: Pcap processing and exploratory analysis. <https://github.com/Noah-213/cerist-http-malicious-dataset>, 2025. Source code and data processing pipeline developed during the CERIST internship.
- [2] Centre de Recherche sur l’Information Scientifique et Technique (CERIST). About cerist, 2024. <https://www.cerist.dz>.
- [3] Roy T. Fielding et al. Http semantics, 2022. RFC 9110, Available at: <https://www.rfc-editor.org/rfc/rfc9110>.
- [4] Kimi Newt. Pyshark: Python wrapper for tshark, 2024. Available at: <https://github.com/KimiNewt/pyshark>.
- [5] NGINX Inc. Nginx http access log module, 2024. Available at: https://nginx.org/en/docs/http/nginx_http_log_module.html.
- [6] OWASP Foundation. Owasp api security top 10, 2023. Available at: <https://owasp.org/www-project-api-security/>.
- [7] Jarry Shaw. Pcapkit: A python pcap file parsing engine, 2024. Available at: <https://github.com/JarryShaw/PyPCAPKit>.